

AKDENİZ UNIVERSITY
DEPARTMENT OF COMPUTER ENGINEERING

CSE321 Introduction to Probability Theory
Midterm Exam

Group A

- You have 45 minutes to solve 25 questions.
- Each question is +4 points. Each incorrect answer costs -1 points.
- Write your all answers to the answer sheet provided.

ID _____ Name _____

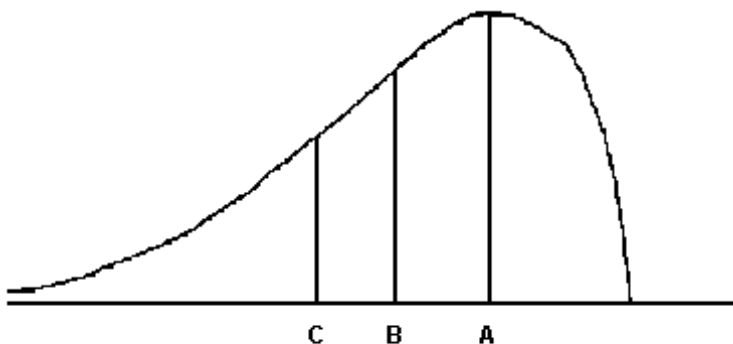
- 1) The overnight shipping business has skyrocketed in the last ten years. The single greatest predictor of a company's success is customer service. A study was conducted to determine the customer satisfaction levels for one overnight shipping business. In addition to the customer's satisfaction level, the customers were asked how often they used overnight shipping. The results are shown below in the following table:

Frequency of Use	Satisfaction level			TOTAL
	High	Medium	Low	
< 2 per month	250	140	10	400
2 - 5 per month	140	55	5	200
> 5 per month	70	25	5	100
TOTAL	460	220	20	700

A customer is chosen at random. Given that the customer uses the company two to five times per month, what is the probability that the customer expressed low satisfaction with the company?

- A) $\frac{1}{40}$ B) $\frac{1}{4}$ C) $\frac{43}{140}$ D) $\frac{1}{140}$

2)



For the distribution drawn here, identify the mean, median, and mode.

- A) A = mode, B = median, C = mean B) A = mode, B = mean, C = median
C) A = mean, B = mode, C = median D) A = median, B = mode, C = mean
- 3) The school newspaper surveyed 100 commuter students and asked two questions. First, students were asked how many courses they were currently enrolled in. Second, the commuter students were asked to estimate how long it took them to drive to campus. Considering these two variables, number of courses would best be considered a _____ variable and drive time would be considered a _____ variable.
- A) discrete; discrete B) discrete; continuous
C) continuous; continuous D) continuous; discrete

- 4) A shoe retailer keeps track of all types of information about sales of newly released shoe styles. One newly released style was marketed to tall people. Listed below are the shoe sizes of 12 randomly selected customers who purchased the new style. Find the mode of the shoe sizes.

$9\frac{1}{2}$ 11 12 $11\frac{1}{2}$
 $8\frac{1}{2}$ $10\frac{1}{2}$ 8 11
 10 11 $9\frac{1}{2}$ 10

- A) $10\frac{1}{4}$ B) $9\frac{1}{2}$ C) 11 D) $10\frac{1}{2}$

- 5) An economy pack of highlighters contains 12 yellow, 6 blue, 4 green, and 3 orange highlighters. An experiment consists of randomly selecting one of the highlighters and recording its color. Find the probability that a blue or yellow highlighter is selected given that a yellow highlighter is selected.

- A) $\frac{1}{2}$ B) 1 C) $\frac{1}{3}$ D) 0

- 6) A fast-food restaurant chain with 700 outlets in the United States has recorded the geographic location of its restaurants in the accompanying table of percentages. One restaurant is to be chosen at random from the 700 to test market a new chicken sandwich.

		Region			
		NE	SE	SW	NW
Population of City	<10,000	8%	6%	3%	0%
	10,000 - 100,000	15%	3%	12%	5%
	>100,000	20%	4%	10%	14%

What is the probability that the restaurant is located in a city with a population over 100,000 and in the southern portion of the United States?

- A) 0.38 B) 0.10 C) 0.04 D) 0.14

- 7) Each manager of a corporation was rated as being either a good, fair, or poor manager by his/her boss. The manager's educational background was also noted. The data appear below:

Manager Rating	Educational Background				Totals
	H. S. Degree	Some College	College Degree	Master's or Ph.D.	
Good	2	3	21	13	39
Fair	8	11	44	24	87
Poor	9	5	7	13	34
Totals	19	19	72	50	160

What is the probability that a randomly chosen manager is either a good managers or has an advanced degree?

- A) $\frac{19}{40}$ B) $\frac{13}{160}$ C) $\frac{89}{160}$ D) $\frac{147}{160}$

- 8) The amount spent on textbooks for the fall term was recorded for a sample of five hundred university students. The mean expenditure was calculated to be \$500 and the median expenditure was calculated to be \$425. Which of the following interpretations of the median is correct?
- A) The average of the textbook costs sampled was \$425
 - B) 50% of the students sampled had textbook costs equal to \$425
 - C) 50% of the students sampled had textbook costs that were less than \$425
 - D) The most frequently occurring textbook cost in the sample was \$425
- 9) A group of 12 people is traveling in three vehicles. If 4 people ride in each vehicle, in how many different ways can the people be assigned to the vehicles?
- A) 11,550
 - B) 11,880
 - C) 34,650
 - D) 495

- 10) A bag of candy was opened and the number of pieces was counted. The results are shown in the table below:

Color	Number
Red	25
Brown	20
Green	20
Blue	15
Yellow	10
Orange	10

Find the probability that a randomly chosen piece of candy is not blue or red.

- A) 0.15
 - B) 0.60
 - C) 0.40
 - D) 0.85
- 11) A number between 1 and 10, inclusive, is randomly chosen. Events A , B , C , and D are defined as follows.
- A : {The number is even}
 - B : {The number is less than 7}
 - C : {The number is less than or equal to 7}
 - D : {The number is 5}
- Identify one pair of independent events.
- A) A and B
 - B) B and D
 - C) A and C
 - D) A and D

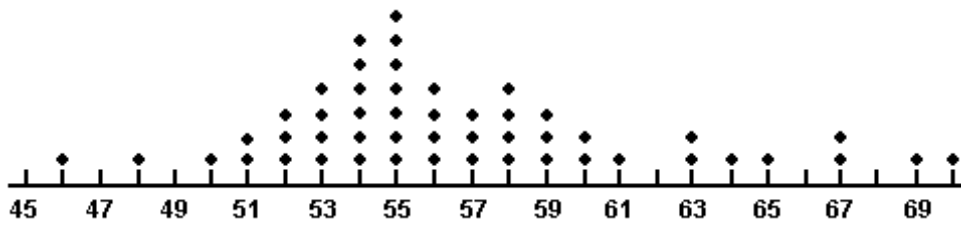
- 12) Which of the following is not an element of descriptive statistical problems?

- A) data are displayed visually in graphs
- B) predictions are made about a larger set of data
- C) information revealed in a data set is summarized
- D) patterns in a data set are identified

- 13) A study in the state of Georgia was conducted to determine the percentage of all community college students who have taken at least one online class. 1500 community college students were contacted and asked if they had taken at least one online class during their time at their community college. These responses were then used to estimate the percentage of all community college students who have taken at least one online class. What type of variable is being collected?

- A) quantitative data
- B) qualitative data

- 14) A dot plot of the speeds of a sample of 50 cars passing a policeman with a radar gun is shown below.



What proportion of the motorists were driving above the posted speed limit of 55 miles per hour?

- A) 0.50 B) 0.14 C) 7 D) 0.64
- 15) Which of the following is not the job of a statistician?
- A) collecting numerical information in the form of data
 - B) determining what information is relevant in a given problem
 - C) implementing new procedures based on the results of a study
 - D) determining whether the conclusions drawn from a study are to be trusted
- 16) Your manager will assign her 9 employees to three tasks, 2 to task A, 4 to task B, and 3 to task C. If she randomly assigns employees to tasks, what is the probability that you and your best friend will be the two assigned to task A?
- A) $\frac{1}{512}$ B) $\frac{1}{36}$ C) $\frac{1}{1260}$ D) $\frac{2}{9}$
- 17) A sociologist recently conducted a survey of citizens over 60 years of age who have net worths too high to qualify for Medicaid but have no private health insurance. The ages of the 25 uninsured senior citizens were as follows:
- 68 73 66 76 86 74 61 89 65 90 69 92 76
62 81 63 68 81 70 73 60 87 75 64 82
- Suppose the mean and standard deviation are 74.04 and 9.75, respectively. If we assume that the distribution of ages is mound-shaped and symmetric, what percentage of the respondents will be between 64.29 and 93.54 years old?
- A) approximately 95%
 - B) approximately 68%
 - C) approximately 84%
 - D) approximately 81.5%
- 18) The speeds of the fastballs thrown by major league baseball pitchers were measured by radar gun. The mean speed was 87 miles per hour. The standard deviation of the speeds was 5 mph. Which of the following speeds would be classified as an outlier?
- A) 95 mph B) 103 mph C) 82 mph D) 77 mph

- 19) A sample of 350 students was selected and each was asked the make of their automobile (foreign or domestic) and their year in college (freshman, sophomore, junior, or senior). The results are shown in the table below.

		Year in College				Total
		Freshman	Sophomore	Junior	Senior	
Car	Foreign	15	65	100	25	205
	Domestic	10	45	80	10	145
	Total	25	110	180	35	350

Which of the following events listed would be considered mutually exclusive events?

- A) The student is a senior and the student drives a domestic automobile.
 - B) The student is a freshman and the student drives a foreign automobile
 - C) The student is a junior and the student drives a domestic automobile
 - D) The student is a junior and the student is a freshman
- 20) A local bakery has determined a probability distribution for the number of cheesecakes it sells in a given day. The distribution is as follows:

Number sold in a day	0	5	10	15	20
Prob (Number sold)	0.21	0.16	0.1	0.24	0.29

Find the number of cheesecakes that this local bakery expects to sell in a day.

- A) 10
 - B) 11.2
 - C) 11.41
 - D) 20
- 21) A survey was conducted to determine how people feel about the quality of programming available on television. Respondents were asked to rate the overall quality from 0 (no quality at all) to 100 (extremely good quality). The stem-and-leaf display of the data is shown below.

Stem	Leaf
3	2 3
4	0 3 4 7 8 9 9 9
5	0 1 1 2 3 4 5
6	1 2 5 6 6
7	4 9
8	
9	9

What percentage of the respondents rated overall television quality as very good (regarded as ratings of 80 and above)?

- A) 9%
 - B) 4%
 - C) 1%
 - D) 36%
- 22) What method of data collection would you use to collect data for a study where a drug was given to 95 patients and a placebo to another group of 95 patients to determine if the drug has an effect on a patient's illness?

- A) survey
- B) designed experiment
- C) observational study
- D) published source

- 23) The table shows the political affiliations and types of jobs for workers in a particular state. Suppose a worker is selected at random within the state and the worker's political affiliation and type of job are noted.

		Political Affiliation		
		Republican	Democrat	Independent
Type of job	White collar	20%	13%	8%
	Blue Collar	17%	18%	24%

Find the probability the worker is not an Independent.

- A) 0.35 B) 0.32 C) 0.68 D) 0.33
- 24) Suppose that B_1 and B_2 are mutually exclusive and complementary events, such that $P(B_1) = .6$ and $P(B_2) = .4$. Consider another event A such that $P(A | B_1) = .2$ and $P(A | B_2) = .5$. Find $P(A)$.
- A) .32 B) .70 C) .88 D) .38
- 25) Suppose that B_1 and B_2 are mutually exclusive and complementary events, such that $P(B_1) = .6$ and $P(B_2) = .4$. Consider another event A such that $P(A | B_1) = .2$ and $P(A | B_2) = .5$. Find $P(B_1 | A)$.
- A) .625 B) .375 C) .800 D) .240